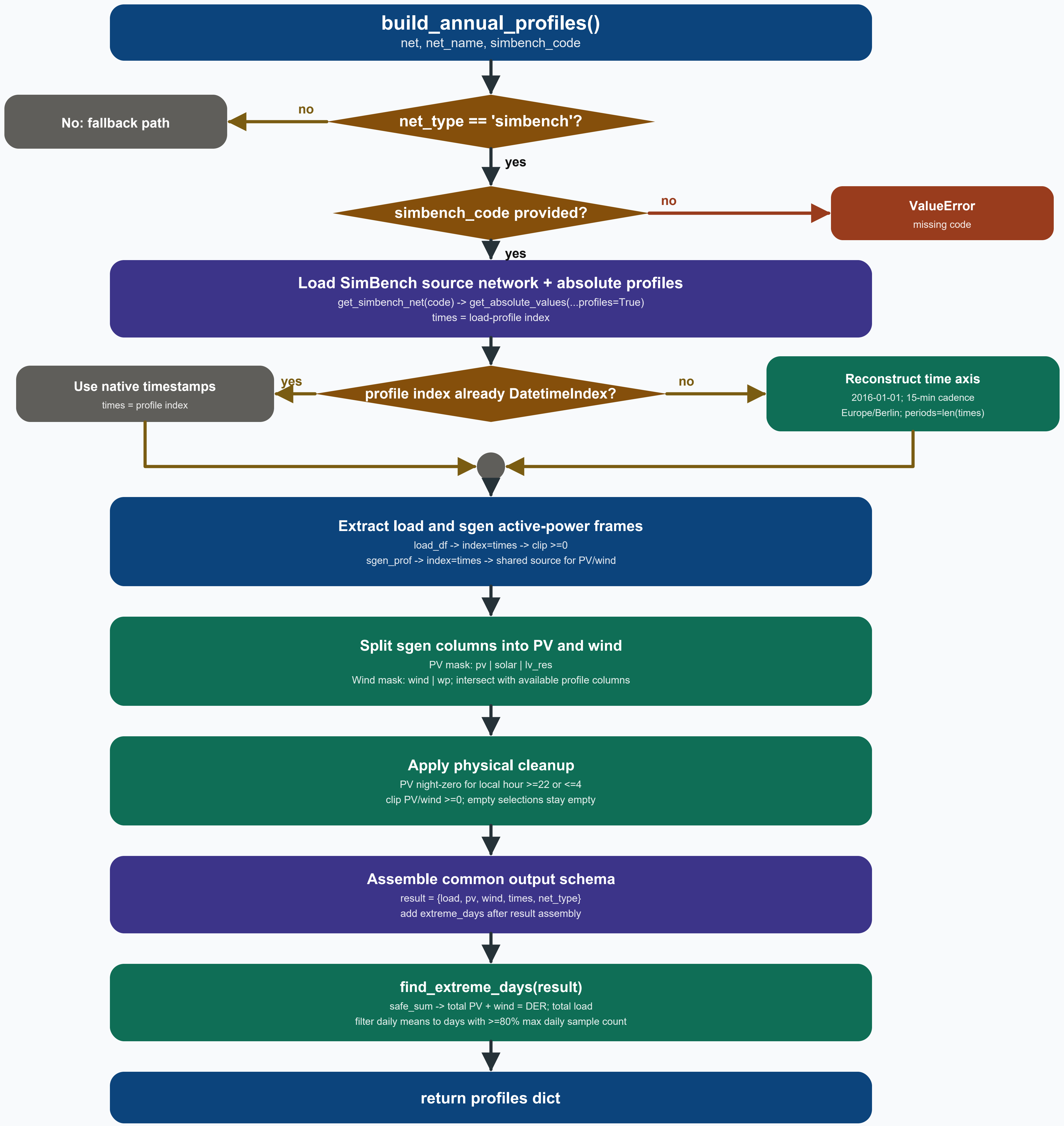

SimBench source + cadence

get_simbench_net(simbench_code) supplies the source network used for profile lookup.

get_absolute_values(...profiles_instead_of_study_cases=True) supplies load/sgen P profiles.

If the returned load-profile index is not a DatetimeIndex, timestamps are reconstructed.

The reconstruction uses 2016-01-01, 15-minute cadence, and Europe/Berlin timezone.

The branch stays at native SimBench resolution; there is no 10-minute resampling here.

DER classification + physical cleanup

PV classification uses net.sgen.type containing pv, solar, or lv_res.

Wind classification uses net.sgen.type containing wind or wp.

Only element indices also present in sgen_prof are retained as profile columns.

PV is forced to zero at local hours >=22 or <=4.

PV and wind are clipped at zero to remove small negative profile-scaling artefacts.

If no matching columns exist, the resulting PV/wind DataFrame is simply empty.

Return schema + extreme-day calculation

Returned keys are load, pv, wind, times, net_type, and extreme_days.

safe_sum() converts a missing or empty profile frame into a zero series on times.

total_der = sum(PV columns) + sum(wind columns); total_load = sum(load columns).

Each series is resampled to daily mean before selecting max/min days.

Days with fewer than 80% of the maximum daily sample count are excluded.

A series with no values or zero maximum yields None for that extreme-day key.

Otherwise max_der, min_der, max_load, and min_load are YYYY-MM-DD strings.